A Common Recovery Transition for Classical and Quantum Information in Random Codes

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(Dated: October 28, 2026)

Reading classical messages requires less than restoring quantum coherence. For random codes under Pauli noise, we show that both tasks have the same critical performance. A finite-size information-spectrum bound sandwiches optimal entanglement fidelity and classical decoding success for every exact unitary 2-design. With independent identical errors, it yields a common Gaussian crossover at positive coding rate, in mean and with high probability. Clifford codes realize the classical optimum by inverse encoding and basis measurement followed by classical decoding.

Recovering a classical label and recovering a quantum state impose different demands on an encoding. A phase error can leave every computational-basis message readable while destroying a superposition. Random quantum encoders spread both kinds of information over a register. Does allowing the receiver to discard coherence then postpone the failure of the code?

We compare these two tasks for the *same* random encoder followed by Pauli noise. The receiver either guesses a uniformly distributed logical-basis message or restores entanglement with a reference. Each receiver is optimized with knowledge of the realized encoder. Our result is that, at positive rate, their optimal performances approach the same Gaussian curve across the coding transition. Thus the easier classical task has no separate threshold or second-order rate advantage within this ensemble, even though its finite-size success probability is larger.

The hashing rate $1 - H(\boldsymbol{p})$ is an established quantum-coding benchmark [1, 2]. Klesse's random-code argument already yields truncated-error-spectrum quantum achievability, including a probability bound over encoders [3]. Second-order quantum achievability [4] and finite-size Clifford analyses [5] quantify attainable rates. Haar-code spectra and coherent information exhibit a hashing transition with a square-root finite-size window [6]; approximate quantum Hamming bounds [7] and stabilizer converses [8] provide complementary constraints. Here a classical decoding converse matches the quantum achievable profile throughout the critical window. Together they determine the optimal classical *and* quantum performance of the same sampled code. The finite-size bracket and its probability bound require only unitary second moments.

One encoder, two tasks.—Let $d = 2^n$ and $M = 2^k \le d$, with integer $k \ge 1$. Embed the logical space by $V_0|j\rangle = |j\rangle|0\rangle^{\otimes(n-k)}$ and encode with $V = UV_0$, where U is drawn from an exact unitary 2-design. Haar and uniform Clifford encoders are examples [9]. The same U encodes all messages; the codewords are orthogonal, not independent random states. A fixed Pauli channel acts after encoding:

$$\mathcal{N}_p(\rho) = \sum_{P\in\{I,X,Y,Z\}^{\otimes n}} p_P P\rho P^\dagger, \quad s(P) = -\log_2 p_P. \tag{1}$$

The distribution is independent of U and may include correlated errors. Define $\mathcal{F}(t) = \mathbb{P}_{P\sim p}[s(P) < t]$. All logarithms are base two.

For $\rho_j = \mathcal{N}_p(V|j\rangle\langle j|V^\dagger)$, the optimal classical success probability is

$$P_{\mathrm{c}}^*(V) = \max_{\{E_j\}} \frac{1}{M}\sum_j \mathrm{tr}(E_j\rho_j). \tag{2}$$

Let $|\Phi_M\rangle = M^{-1/2}\sum_j |j\rangle|j\rangle$ and $\mathcal{V}(\rho) = V\rho V^\dagger$. The optimal entanglement fidelity is

$$F_{\mathrm{q}}^*(V) = \max_{\mathcal{D}} \langle\Phi_M|[\mathrm{id}\otimes(\mathcal{D}\mathcal{N}_p\mathcal{V})](\Phi_M)|\Phi_M\rangle, \tag{3}$$

where $\mathcal{D}$ is any quantum channel back to the logical space. Fidelity here is the overlap, not its square root. Both optimizations precede averaging over U. Encoding and decoding are ideal; neither a noise record nor access to the environment is supplied.

A common finite-size bracket.—The following bounds compare the two optimized tasks without identifying their decoders.

Theorem 1. *For any distribution in Eq. (1), any exact unitary 2-design encoder, and every $a > 0$,*

$$\begin{aligned}\mathcal{F}(n-k-a) - 8\,2^{-a} \le \mathbb{E}F_{\mathrm{q}}^* \le \mathbb{E}P_{\mathrm{c}}^* \\ \le \mathcal{F}(n-k+2a) + \frac{5}{4}\,2^{-a} + \frac{2^a}{4M}.\end{aligned} \tag{4}$$

Moreover, $F_{\mathrm{q}}^*(V) \le P_{\mathrm{c}}^*(V)$ for every encoder. With $F_- = \mathcal{F}(n-k-a)$ and $F_+ = \mathcal{F}(n-k+2a)$, for any $\eta > 0$,

$$\begin{aligned}\mathbb{P}_U[F_{\mathrm{q}}^* < F_- - \eta \text{ or } P_{\mathrm{c}}^* > F_+ + \eta] \\ \le \frac{37\,2^{-a}}{4\eta} + \frac{2^a}{4M\eta}.\end{aligned} \tag{5}$$

The relevant spectral budget is $n-k$. Likely errors consume physical space that would otherwise hold logical messages. Quantum recovery is possible for a sufficiently small likely-error family; the classical converse shows that abandoning phase information does not recover the spectral mass outside the same budget. The proof controls overlaps in aggregate and does not replace error images by independent or exactly orthogonal vectors.

Proof.—For a recovered logical channel with Kraus operators A_ℓ,

$$F_e = \frac{1}{M^2}\sum_\ell |\operatorname{tr} A_\ell|^2 \le \frac{1}{M}\sum_{j,\ell}|(A_\ell)_{jj}|^2 \le P_{\mathrm{c}}^*. \tag{6}$$

The middle expression is the success of that recovery followed by a logical-basis measurement. This proves the pointwise ordering.

For the lower bound, take $\mathcal{G}=\{s<n-k-a\}$, $L=|\mathcal{G}|<d2^{-a}/M$, and define

$$S=\sum_{P\in\mathcal{G}} PVV^\dagger P^\dagger, \qquad R_P = V^\dagger P^\dagger S^{-1/2}. \tag{7}$$

The inverse is on the support of S. The R_P obey $\sum_P R_P^\dagger R_P = \Pi_{\mathrm{supp}S}$; completing them on its kernel gives a trace-preserving square-root recovery [10]. For $v_{jP}=PV|j\rangle$ and $a_{jP}=\langle v_{jP}|S^{-1/2}|v_{jP}\rangle$, Hayashi–Nagaoka [11] gives

$$1-a_{jP}^2 \le 4\sum_{(i,Q)\neq(j,P)}|\langle v_{jP}|v_{iQ}\rangle|^2 =: 4C_{jP}. \tag{8}$$

Also $0\le a_{jP}\le 1$. Keeping the terms where the actual error matches the recovery Kraus index in the fidelity sum yields

$$\begin{aligned} F_e &\ge \sum_{P\in\mathcal{G}} p_P\left(\frac{1}{M}\sum_j a_{jP}\right)^2 \ge p(\mathcal{G}) - 8X,\\ X &= \sum_{P\in\mathcal{G}}\frac{p_P}{M}\sum_j C_{jP}. \end{aligned} \tag{9}$$

The label P in Eq. (7) is an internal Kraus index, not a known noise realization. Exact unitary second moments and orthogonality of the code columns give

$$\mathbb{E}C_{jP} = \frac{(L-1)(dM-1)}{d^2-1} \le \frac{ML}{d}, \tag{10}$$

so $\mathbb{E}X \le 2^{-a}$. This proves the lower bound, including nonuniform error probabilities. The empty $\mathcal{G}$ case is trivial.

For the upper bound, every POVM and every $c>0$ satisfy

$$P_{\mathrm{succ}} \le \frac{cd}{M} + \frac{1}{M}\sum_j \operatorname{tr}(\rho_j - cI)_+. \tag{11}$$

Split each output at error probability c' into a head and a tail B_j. The head contributes at most its mass, while $(\lambda-c)_+\le\lambda^2/(4c)$ bounds the tail. Each code column is a state 2-design, and Pauli orthogonality gives

$$\mathbb{E}\operatorname{tr}B_j^2 = \frac{dv_t+m_t^2}{d+1} \le c' + \frac{1}{d}, \tag{12}$$

where $m_t=\sum_{p_P\le c'}p_P$ and $v_t=\sum_{p_P\le c'}p_P^2 \le c'm_t$. Putting $\gamma=cd$ and $c'=c2^{-a}$ gives

$$\mathbb{E}P_{\mathrm{c}}^* \le \frac{\gamma}{M}+\frac{1}{4\gamma}+\frac{2^{-a}}{4}+\mathcal{F}(n-\log_2\gamma+a). \tag{13}$$

Choose $\gamma=M2^{-a}$. The same pointwise constructions control the positive deviations from F_- and F_+; Markov's inequality gives Eq. (5). Full details appear in the Supplemental Material (SM) [12].

The shared critical curve.—For independent identical errors, $p=\boldsymbol{p}^{\otimes n}$, write $H=\mathbb{E}[-\log_2 p_\alpha]$ and $\sigma^2=\mathrm{Var}[-\log_2 p_\alpha]$ for one qubit. At fixed positive rate $R=k/n$, Eq. (4) gives exponential approach to one for $R<1-H$, and exponential decay to zero for $R>\max(1-H,0)$. These are ensemble-specific coding thresholds.

More precisely, fix $\boldsymbol{p}$ with $0<H<1$ and $\sigma>0$, and let

$$k_n = n(1-H)+\sqrt{n}\,\sigma z + O(1) \tag{14}$$

for fixed z. Berry–Esseen [13], applied to the two CDFs in Eq. (4) with $a=2\log_2 n$, yields

$$\boxed{\begin{aligned}\mathbb{E}F_{\mathrm{q}}^* &= \Phi(-z)+O\left(\frac{\log n}{\sqrt{n}}\right),\\ \mathbb{E}P_{\mathrm{c}}^* &= \Phi(-z)+O\left(\frac{\log n}{\sqrt{n}}\right).\end{aligned}} \tag{15}$$

Here Φ is the normal CDF. Equation (5) with $\eta=n^{-1/2}$ shows that both quantities follow this curve for the same encoder outside an exceptional set of probability at most $37/(4n^{3/2})+n^{5/2}/(4M_n)$. The constants are uniform over the choice of exact 2-design. At a fixed target error ϵ, the classical and quantum ensemble coding sizes therefore both obey

$$k_\epsilon = n(1-H)+\sqrt{n}\,\sigma\Phi^{-1}(\epsilon)+O(\log n). \tag{16}$$

Errors are respectively $1-P_{\mathrm{c}}^*$ and $1-F_{\mathrm{q}}^*$. Thus equality extends through the second-order term, not just the hashing rate. Fixed message dimension, zero surprisal variance, and the endpoint $H=1$ are outside this Gaussian statement; Eq. (4) remains valid there.

Exact Clifford decoding.—For a Clifford encoder, conjugating a physical error through $U^\dagger$ produces logical Pauli labels (x,z) and an auxiliary bit-flip syndrome s. Let $w(s,x,z)$ include the probability of all physical errors

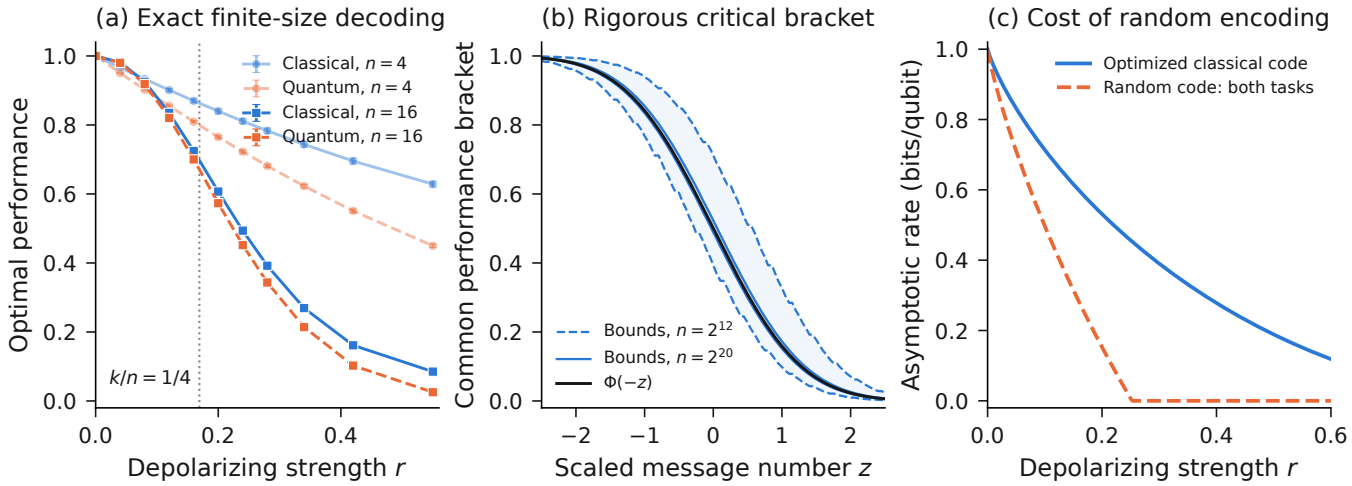


FIG. 1. Two optimal tasks and one asymptotic transition. (a) Exact classical success and entanglement fidelity for uniform Clifford encoders with $k/n = 1/4$; 64 encoders per size, reused across noise strengths. Markers show encoder means and one-standard-error bars; lines join points. The vertical line solves $H(1 - 3r/4, r/4, r/4, r/4) = 3/4$. The finite-size classical advantage is resolved. (b) Rigorous bounds in Eq. (4) for $r = 0.1$, $a = 2\log_2 n$, and k chosen by Eq. (14), compared with $\Phi(-z)$. These are binomial-CDF evaluations, not large-system decoding simulations. (c) The common random-code rate $[1 - H]_+$ compared with the optimized classical capacity $1 - h_2(r/2)$ of the same depolarizing channel.

with these labels. Auxiliary phase errors are summed. Orthogonal syndrome sectors reduce optimal quantum recovery to correction of the most likely logical Pauli [8], while phase labels are irrelevant to a classical basis message. Consequently,

$$F_{\mathrm{q}}^{*} = \sum_{s} \max_{x,z} w(s,x,z),$$
$$P_{\mathrm{c}}^{*} = \sum_{s} \max_{x} \sum_{z} w(s,x,z). \tag{17}$$

The classical outputs commute. Inverse Clifford encoding, computational-basis measurement, and classical maximum-likelihood processing attain P_{c}^{*}. The processing is not guaranteed to be efficient. Equation (17) also makes the finite-size advantage transparent: classical decoding can combine errors that differ only in their logical phase. The theorem states that this advantage does not survive at leading order inside the regular critical window.

We evaluate Eq. (17) for $n = 4, 8, 12, 16$, using an exact binary Fourier sum over syndrome and logical labels. No independent-vector or random-matrix replacement is made. Figure 1(a) shows the smallest and largest sizes. Independent direct Pauli enumeration and primal–dual semidefinite optimization at $n \leq 3$ check both scores and the explicit recovery construction; methods, all sizes, seeds, and raw encoder values accompany the SM. The modest simulated sizes illustrate finite-size behavior; the shared asymptotic curve follows from the proof and its spectral bounds [Fig. 1(b)].

What random encoding costs.—The common rate is a property of random quantum encoders, not an equality of classical and quantum channel capacities. For $\mathcal{N}_r(\rho) = (1-r)\rho + rI/2$, classical product-state coding achieves the capacity $1 - h_2(r/2)$ [14]. At $r = 0.1$ it is 0.714 bits per qubit, whereas the common random-code rate is 0.497. The latter vanishes at $r_c = 0.252386\ldots$, where the optimized classical capacity is still 0.453 [Fig. 1(c)]. Thus the noise continues to support classical communication after these random encodings lose every positive rate. The receiver's permission to discard coherence does not recover the encoding advantage of a code designed specifically for classical messages.

Random-state contraction [15] and coherent information [6] probe related aspects of noisy encodings. Liu *et al.* also bound accessible information for a full state-2-design alphabet [16]. Here the alphabet is the M orthogonal columns at fixed, known U: the receiver optimizes separately for that alphabet before we average over U. A classical converse for this conditional codebook matches quantum achievability, explaining the common recovery curve. Extending this result to approximate designs and noise accumulated during encoding would connect it to finite-depth devices.

Acknowledgments.—Generative AI was used substantively for problem selection, derivations, literature searches, code, figures, and manuscript preparation, including independent model-assisted review. Tools included Claude Code with Claude Opus 5.5 and the session model identified as `mythos-magpie-2161436569`; an earlier external GPT review had an unrecorded model version. The author is responsible for the submitted work. The new recovery theorem has not been formalized in Lean; an earlier conditional two-state formalization is

retained separately.

Data availability.—Source, raw encoder samples, numerical verification records, and figure data are supplied with this manuscript. No public archive identifier has yet been assigned.

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[12] See Supplemental Material for the full recovery proof, probability bounds, second-order inversion, Clifford optimality, and numerical methods.

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